

Conditional Sharp Partial Identification of Diversification Histories

Affine Measure Geometry, Event Congruence, Certified Extremes and a Confirmatory CRABS Comparison

Anonymous agentic research candidate

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Abstract

Extant timetrees identify a pulled diversification signal but do not generally identify time-varying speciation and extinction rates separately. Existing methods therefore construct or sample alternative histories in a congruence class. Conditional on a fixed pulled speciation rate, we represent homogeneous birth–death histories by a cumulative-loss measure beneath an explicit survival barrier. Reciprocal speciation is affine in this measure, and its Radon–Nikodym derivative with respect to pulled scale is the turnover ratio μ/λ . We prove finite-event congruence under stated fixed-stem conditioning, sharp turnover-capped bounds for every finite cap, explicit finite-constraint envelopes and infeasibility certificates, and an exact endpoint-sampling law. We then test whether certification adds decision utility to CRABS exploration in a prospectively frozen 1,100-cell benchmark. All 1,944 deterministic validation cells passed. Finite CRABS rejection exploration left a normalized endpoint deficit above 0.05 in all 300 primary cells, including 60 cells prospectively treated as structural censors after a documented nontermination amendment; a boundary constructor reached both discrete endpoints in all 300 cells. Certification changed 532 of 3,840 evaluable decision-query statuses (13.85%; descriptive Wilson range 12.80–14.98%), below the frozen 20% H4 criterion, with zero false certificates. Clean-install, normal/optimized-task and runtime gates passed. Exact certification therefore exposed systematic endpoint incompleteness but did not meet the prospectively frozen essential-complement criterion. A public mammal curve remains a plug-in sensitivity illustration only. The contribution is exact conditional projection and a reproducible diagnostic complement, not a replacement for likelihoods, identifiable restricted models or fossilised birth–death inference.

Assurance boundary. This is an unrefereed theorem-led candidate with a completed internal confirmatory benchmark. Internal replay, independent linear-programming checks, direct branching-process simulation, deliberate negative controls and the sealed affine–CRABS ledger pass their stated integrity checks. The failed H4 utility gate is retained as the primary decision and prohibits “must-have” or essential-complement language. External specialist review of the fixed-stem event theorem, a systematic priority search, simultaneous uncertainty for the pulled signal, a fossil observation bridge and independent external replay remain open. “Sharp” always means sharp over the stated conditional model class.

Persistent identifier. The prior 0.2.1 candidate is archived at [10.5281/zenodo.21851319](https://doi.org/10.5281/zenodo.21851319). This 0.3.0-candidate review draft is not yet archived and must not be attributed to that specific-version DOI.

1 Decision problem and contribution

Let $\tau \in [0, T]$ denote age before an analysis origin, with $\tau = 0$ at that origin and larger τ further in the past. A homogeneous time-varying birth–death history has speciation rate $\lambda(\tau) > 0$,

extinction rate $\mu(\tau) \geq 0$, and sampling probability $\rho \in (0, 1]$ at the analysis origin. If $E(\tau)$ is the probability that a lineage at age τ leaves no sampled descendant, define

$$u(\tau) = 1 - E(\tau), \quad \lambda_p(\tau) = \lambda(\tau)u(\tau). \quad (1)$$

The pulled speciation rate λ_p is identified by the reconstructed extant-timetre likelihood under the generalized homogeneous model, whereas λ and μ are not separately identified without further restrictions (Louca and Pennell 2020; Helmstetter et al. 2022).

Existing tools construct or sample histories in the resulting congruence class (Höhna et al. 2022; Andréoletti and Morlon 2023; Kopperud et al. 2023). They answer an exploration question. The target here is a partial-identification question:

For a fixed pulled signal and explicit restrictions, what is the complete feasible trajectory set, what are the sharp extrema of a stated target, and which constraints explain infeasibility?

The paper contributes five linked results. First, it identifies an affine positive-measure coordinate for the smooth fibre and a Stieltjes completion containing incomplete sampling and deterministic survival events. Second, it secures the finite-event result under a precisely stated fixed-stem conditioning convention. Third, it solves turnover-capped pointwise and finite-constraint identification, including the previously omitted $c > 1$ regime. Fourth, it quantifies the failure of a transparent finite sampler to attain exact endpoints. Fifth, it subjects the certification claim to a prospectively frozen direct comparison with CRABS and retains the failed primary utility gate.

2 Smooth affine fibre

Assume that λ_p is continuous and strictly positive and define

$$F(\tau) = \exp\left\{\int_0^\tau \lambda_p(s) \, ds\right\}, \quad F(0) = 1, \quad F' = \lambda_p F. \quad (2)$$

The survival probability satisfies

$$u' = \lambda_p(1 - u) - \mu u, \quad u(0) = \rho. \quad (3)$$

Set

$$A(\tau) = F(\tau)[1 - u(\tau)]. \quad (4)$$

Theorem 1 (Smooth affine-fibre representation). *Fix $\lambda_p > 0$ and F from (2). The map*

$$(\rho, \lambda, \mu) \mapsto A = F\left(1 - \frac{\lambda_p}{\lambda}\right)$$

is a bijection between continuously differentiable histories with pulled rate λ_p and functions $A \in C^1([0, T])$ satisfying

$$0 \leq A(0) < 1, \quad A' \geq 0, \quad A < F.$$

The inverse is

$$\rho = 1 - A(0), \quad u = \frac{F - A}{F}, \quad \lambda = \frac{\lambda_p F}{F - A}, \quad \mu = \frac{A'}{F - A}. \quad (5)$$

If ν is the positive measure with cumulative function A , then

$$\nu = (1 - \rho)\delta_0 + A'(\tau) \, d\tau, \quad \frac{d\nu_{\text{ac}}}{dF} = \frac{\mu}{\lambda} \quad \text{almost everywhere.} \quad (6)$$

Proof. Differentiate (4), use $F' = \lambda_p F$ and substitute (3):

$$A' = \lambda_p F(1 - u) - F\{\lambda_p(1 - u) - \mu u\} = \mu u F \geq 0.$$

The initial value is $A(0) = 1 - \rho$, and $A < F$ is equivalent to $u > 0$. Conversely, (5) defines positive u and λ and non-negative μ ; direct substitution recovers (3) and $\lambda_p = \lambda u$. Finally,

$$\frac{d\nu_{ac}}{dF} = \frac{A'}{F'} = \frac{\mu u F}{\lambda_p F} = \frac{\mu}{\lambda}.$$

□

At each age,

$$\frac{1}{\lambda(\tau)} = \frac{1}{\lambda_p(\tau)} - \frac{A(\tau)}{\lambda_p(\tau)F(\tau)}, \quad (7)$$

so reciprocal speciation is affine in ν . The admissible fibre

$$\mathcal{V}_F = \{\nu \in \mathcal{M}_+([0, T]) : \nu([0, \tau]) < F(\tau) \ \forall \tau\} \quad (8)$$

is convex and closed under pointwise meet and join of cumulative functions. It is not a cone because positive scaling can cross the barrier. Larger cumulative loss means larger λ , smaller u and smaller $1/\lambda$.

3 Measure completion and finite survival events

For a finite positive Radon measure $\nu \in \mathcal{V}_F$, let

$$A(\tau) = \nu([0, \tau]), \quad q(\tau) = F(\tau) - A(\tau), \quad u = q/F. \quad (9)$$

Use right-continuous cumulative functions and let $q_-(\tau) = F(\tau) - A(\tau^-)$. Define

$$dK(\tau) = \frac{d\nu(\tau)}{q_-(\tau)}. \quad (10)$$

Proposition 2 (Stieltjes loss equation). *The survival function solves*

$$du = \lambda_p(1 - u) d\tau - u_- dK. \quad (11)$$

On the absolutely continuous part, $dK/d\tau = \mu$. If ν has an atom m at age a , then

$$u(a+) = s_a u(a-), \quad q(a+) = s_a q(a-), \quad m = (1 - s_a)q(a-), \quad (12)$$

where $s_a = 1 - m/q(a-) \in (0, 1]$.

Proof. Since $u = 1 - A/F$ and F is continuous,

$$du = \lambda_p(1 - u) d\tau - F^{-1} dA.$$

Because $q_- = Fu_-$ and $dA = d\nu = q_- dK$, this is (11). The strict barrier gives $m < q(a-)$; taking jumps yields (12). □

An atom at zero represents incomplete sampling. An interior atom represents independent deterministic survival of contemporaneous lineages. It is not a value of the continuous ratio μ/λ .

3.1 Descendant-count law

Let N_τ be the number of sampled descendants at the analysis origin from one lineage at age τ , with probability-generating function

$$G_\tau(z) = \mathbb{E}[z^{N_\tau}] = \sum_{k \geq 0} p_k(\tau) z^k.$$

Between events,

$$\partial_\tau G = \lambda(G^2 - G) + \mu(1 - G), \quad G_0(z) = 1 - \rho + \rho z. \quad (13)$$

At an event with survival s_a , moving from the younger to the older side,

$$G_{a+}(z) = 1 - s_a + s_a G_{a-}(z). \quad (14)$$

Theorem 3 (Zero-inflated geometric law with finite events). *For an absolutely continuous history with finitely many deterministic survival events,*

$$G_\tau(z) = 1 - u(\tau) + u(\tau) \frac{a(\tau)z}{1 - [1 - a(\tau)]z}, \quad a(\tau) = \frac{1}{F(\tau)}. \quad (15)$$

Thus

$$p_0 = 1 - u, \quad p_k = \frac{u}{F} \left(1 - \frac{1}{F}\right)^{k-1}, \quad k \geq 1, \quad (16)$$

and

$$\frac{p_1}{u} = \frac{1}{F}, \quad \Pr(N_\tau = k \mid N_\tau > 0) = \frac{1}{F} \left(1 - \frac{1}{F}\right)^{k-1}. \quad (17)$$

Finite events change u but not the conditional positive descendant-count law.

Proof. At $\tau = 0$, $a = 1$ and (15) equals $1 - \rho + \rho z$. Between events, $a' = -\lambda_p a = -\lambda u a$. Differentiate (15), substitute $u' = \lambda u(1 - u) - \mu u$ and $a' = -\lambda u a$, and recover (13). Uniqueness of the backward equation proves the form between events. Equation (14) multiplies u by s_a without changing $a = 1/F$, so the form persists across every event. Expanding the geometric series gives (16)–(17). $\square$

3.2 Complete fixed-stem reconstructed-tree law

Fix a stem age T . Let $0 < t_1 < \dots < t_{n-1} < T$ be the ordered internal-node ages after marginalising over the exchangeable ranked topology. Conditional on the ages, the choice of splitting lineage and daughter orientation has a topology law that depends on n and the reporting convention, but not on the rate history. Define

$$h(t) = \lambda(t)p_1(t) = \frac{\lambda_p(t)}{F(t)} = -\frac{d}{dt} \frac{1}{F(t)}, \quad (18)$$

so $\int_0^T h(t) dt = 1 - 1/F(T)$.

Theorem 4 (Finite-event congruence at fixed stem age). *Under theorem 3, the unconditioned joint density of $N_T = n$ and the ordered internal-node ages, with ranked topology marginalised, is*

$$g_{n,T}(t_1, \dots, t_{n-1}) = p_1(T) \prod_{i=1}^{n-1} i \lambda(t_i) p_1(t_i). \quad (19)$$

It integrates to $p_n(T)$. Conditional on stem survival,

$$f_{n,T}^{\text{surv}}(t_1, \dots, t_{n-1}) = \frac{(n-1)!}{F(T)} \prod_{i=1}^{n-1} \frac{\lambda_p(t_i)}{F(t_i)}, \quad (20)$$

and conditional additionally on $N_T = n$,

$$f_T^n(t_1, \dots, t_{n-1}) = \frac{(n-1)!}{[1 - 1/F(T)]^{n-1}} \prod_{i=1}^{n-1} \frac{\lambda_p(t_i)}{F(t_i)}. \quad (21)$$

Conditional on these ages, homogeneous branching and independent event thinning make the ranked-topology law exchangeable and independent of the rate history. Hence histories sharing λ_p have the same complete fixed-stem reconstructed-tree law under either conditioning, including histories with finitely many independent deterministic survival events.

Proof. The branching property gives (19) for the topology-marginal ordered ages. Before the first observed split the stem contributes $p_1(T)$. With i reconstructed lineages, summing over which lineage generates the next observed split produces $i\lambda(t_i)$, and the new observed branch contributes $p_1(t_i)$. Conditional on that split, exchangeability makes the lineage choice uniform among the i lineages; daughter orientation is likewise rate-independent. Independent event thinning preserves exchangeability and the recursion. By (16) and (18),

$$g_{n,T} = \frac{u(T)}{F(T)} (n-1)! \prod_{i=1}^{n-1} h(t_i).$$

Integration over the ordered simplex yields

$$\frac{u(T)}{F(T)} (n-1)! \frac{1}{(n-1)!} \left\{ \int_0^T h(t) dt \right\}^{n-1} = \frac{u(T)}{F(T)} \left(1 - \frac{1}{F(T)} \right)^{n-1} = p_n(T).$$

Dividing by $u(T)$ proves (20); dividing by the conditional geometric probability in (17) proves (21). $\square$

Remark 5 (Conditioning scope). Theorem 4 fixes stem age and covers stem-survival and observed-tip-count conditioning. It does not claim equality under random-origin priors, crown conditioning or other normalisers. Those cases require separate derivations. Event-aware reconstructed-process and coalescent point-process results remain the process-theoretic antecedents (Lambert and Stadler 2013; Höhna 2015; MacPherson et al. 2022).

4 Turnover caps and the phase transition at $c = 1$

In pulled-scale time $x = F(\tau)$, set $n(x) = A(F^{-1}(x))$ and $q(x) = x - n(x)$. For absolutely continuous histories,

$$n'(x) = \frac{\mu}{\lambda} =: \varepsilon(x), \quad q'(x) = 1 - \varepsilon(x), \quad n(1) = 1 - \rho, \quad q(1) = \rho. \quad (22)$$

Assume no interior atoms and impose $0 \leq \varepsilon \leq c$ for a finite $c \geq 0$.

Theorem 6 (Sharp pointwise bounds for every finite turnover cap). *At any pulled scale $x \geq 1$,*

$$q(x) \leq \rho + x - 1, \quad (23)$$

and

$$\inf q(x) = \max\{\rho + (1 - c)(x - 1), 0\}. \quad (24)$$

The upper endpoint is attained by $\varepsilon = 0$. The lower endpoint is attained by $\varepsilon = c$ when $\rho + (1 - c)(x - 1) > 0$; otherwise zero is an unattained infimum because $q > 0$. Consequently,

$$\lambda_{\min}(x) = \frac{\lambda_p(x)x}{\rho + x - 1}, \quad (25)$$

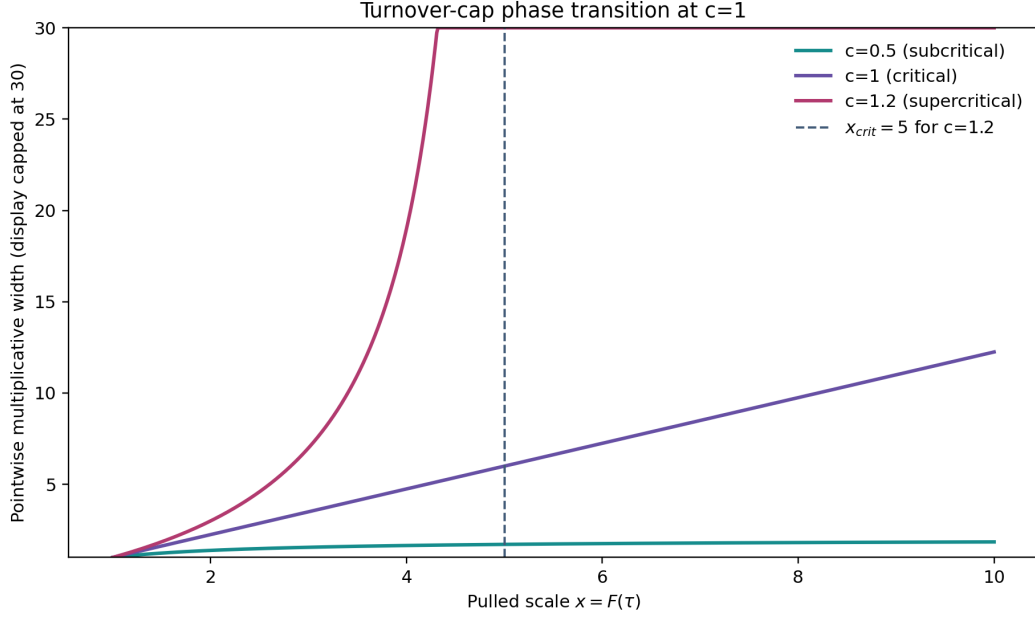


Figure 1: Pointwise identification width for $\rho = 0.8$. For $c < 1$ the width has a finite global limit; at $c = 1$ it grows without a global bound; for $c > 1$ it diverges at the vertical critical pulled scale. The displayed width is capped at 30. These are conditional mathematical bounds, not empirical estimates.

while

$$\sup \lambda(x) = \begin{cases} \frac{\lambda_p(x)x}{\rho + (1-c)(x-1)}, & \rho + (1-c)(x-1) > 0, \\ +\infty, & \rho + (1-c)(x-1) \leq 0. \end{cases} \quad (26)$$

Proof. Integrating $1 - c \leq q' \leq 1$ from $x = 1$ gives the candidate endpoints while they remain positive. The upper endpoint follows from $\varepsilon = 0$. If the lower affine expression is positive, $\varepsilon = c$ attains it. If it is non-positive, choose a constant $\varepsilon \leq c$ that gives any prescribed $q(x) = \delta > 0$; along the linear path, q remains positive. Letting $\delta \downarrow 0$ proves the infimum. Since $\lambda = \lambda_p x / q$, the rate bounds follow. $\square$

Corollary 7 (Three cap regimes). *The identification geometry has three regimes.*

(i) If $c < 1$, the multiplicative width obeys

$$\frac{\lambda_{\max}}{\lambda_{\min}} = \frac{\rho + x - 1}{\rho + (1-c)(x-1)} \leq \frac{1}{1-c}. \quad (27)$$

(ii) If $c = 1$, both pointwise endpoints are finite and attained for every finite x , but the width $(\rho + x - 1)/\rho$ has no uniform deep-time bound.

(iii) If $c > 1$, define

$$x_{\text{crit}} = 1 + \frac{\rho}{c-1}. \quad (28)$$

The speciation supremum is finite below x_{crit} and infinite at and beyond it.

The familiar assumption $c \leq 1$ means $\mu \leq \lambda$ at every non-event time and therefore excludes continuous negative net diversification. The extension to $c > 1$ makes the cost of allowing decline explicit rather than hiding it in a sampler.

5 Finite constraints: exact envelopes and certificates

Suppose external or structural information gives

$$L_i \leq n(x_i) \leq U_i, \quad 1 \leq x_0 < \dots < x_m. \quad (29)$$

The present anchor is normally included as $x_0 = 1$ and $L_0 = U_0 = 1 - \rho$.

Theorem 8 (Directed-Lipschitz feasibility and sharp envelopes). *There exists a non-decreasing, c -Lipschitz function satisfying (29) if and only if*

$$L_i \leq U_j + c(x_i - x_j)_+ \quad \text{for every } i, j. \quad (30)$$

When feasible, the least and greatest such functions are

$$\underline{n}(x) = \max_i \{L_i - c(x_i - x)_+\}, \quad (31)$$

$$\bar{n}(x) = \min_i \{U_i + c(x - x_i)_+\}. \quad (32)$$

Every feasible n obeys $\underline{n} \leq n \leq \bar{n}$, and both envelopes are feasible. A violating pair (i, j) is a complete finite certificate of generic infeasibility.

Proof. Monotonicity and the slope cap imply

$$n(x) \geq L_i - c(x_i - x)_+, \quad n(x) \leq U_i + c(x - x_i)_+.$$

Taking the maximum and minimum gives the necessary bounds and (30). Each lower cone is non-decreasing and c -Lipschitz, and pointwise maxima preserve both properties. Under (30), $\underline{n}(x_j) \leq U_j$, while construction gives $\underline{n}(x_j) \geq L_j$; hence $\underline{n}$ is feasible and least. The upper proof is symmetric. $\square$

Corollary 9 (Positive-survival barrier). *Let the domain be $[1, X]$ and include the anchor $n(1) = 1 - \rho$. For $c \leq 1$, generic feasibility in theorem 8 automatically implies $n(x) < x$. For $c > 1$, a positive-survival history exists if and only if (30) holds and*

$$L_i < x_i \quad \text{for every } i. \quad (33)$$

In that case the least envelope itself respects the barrier.

Proof. For $c \leq 1$, the anchor and $n' \leq c$ give $n(x) \leq 1 - \rho + c(x - 1) < x$. For $c > 1$, consider one lower cone $\phi_i(x) = L_i - c(x_i - x)_+$. The function $\phi_i(x) - x$ increases up to x_i and decreases afterwards, so its maximum is $L_i - x_i$. Hence, if every $L_i < x_i$, the least envelope $\underline{n} = \max_i \phi_i$ satisfies $\underline{n}(x) < x$ throughout the domain and supplies an admissible feasible history. Conversely, if $L_i \geq x_i$ for some i , every function satisfying that lower constraint has $n(x_i) \geq L_i \geq x_i$ and violates positive survival at x_i . $\square$

Corollary 10 (Minimum compatible turnover cap). *If $L_i \leq U_j$ whenever $x_i \leq x_j$, the smallest generic cap is*

$$c_\star = \max_{x_i > x_j} \frac{(L_i - U_j)_+}{x_i - x_j}. \quad (34)$$

If some $x_i \leq x_j$ has $L_i > U_j$, monotonicity alone makes the system infeasible for every finite cap. A result $c_\star > 1$ means that every absolutely continuous solution requires some negative-net-diversification interval, unless the data bridge, event structure or other assumptions change.

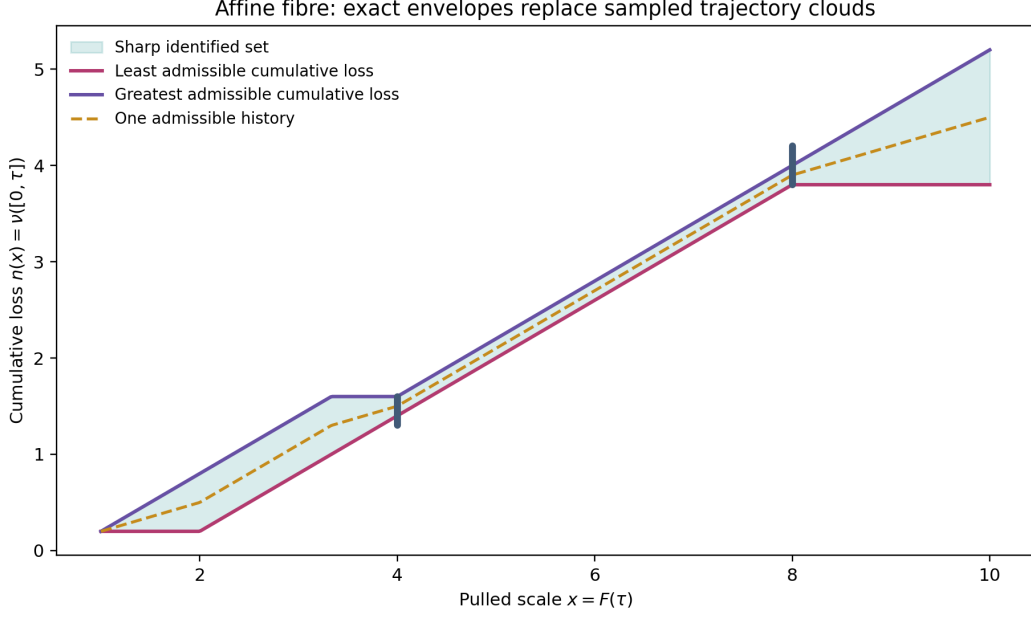


Figure 2: Least and greatest feasible cumulative-loss trajectories under finite interval constraints. The shading is the pointwise projection between two feasible extremal functions; it is not a licence to choose values independently at each x .

Corollary 11 (Order-monotone targets). *Let $\Psi[n]$ be increasing under pointwise cumulative-loss order. Over a feasible set from theorem 8,*

$$\inf \Psi = \Psi[\underline{n}], \quad \sup \Psi = \Psi[\bar{n}],$$

with reversed roles for decreasing targets. This covers point evaluations of n , A , λ and the deterministic diversity quantity defined below, and positive weighted integrals of order-monotone transforms. Non-monotone targets, including peak timing and some path extrema, require a separate optimisation problem.

The two envelopes describe the pointwise projection of the functional identified set. An arbitrary curve drawn inside the shaded band need not satisfy the cross-time constraints.

6 Why finite random sampling is not certification

The distinction between exploration and certification can be quantified without criticising any particular implementation. Consider an equal pulled-scale grid $x_j = 1 + j\Delta$, $j = 0, \dots, m$, and independently draw each interval turnover $\varepsilon_j \sim \text{Uniform}(0, c)$. The sharp maximal-loss endpoint is

$$n_{\max}(x_m) = 1 - \rho + mc\Delta. \quad (35)$$

A sampled endpoint has deficit

$$D = n_{\max}(x_m) - n(x_m) = \Delta \sum_{j=1}^m (c - \varepsilon_j). \quad (36)$$

Theorem 12 (Factorial endpoint-sampling law). *Under the matched sampler above, $\Pr(D = 0) = 0$. For $0 \leq \eta \leq c\Delta$,*

$$p_\eta := \Pr(D \leq \eta) = \frac{1}{m!} \left(\frac{\eta}{c\Delta} \right)^m. \quad (37)$$

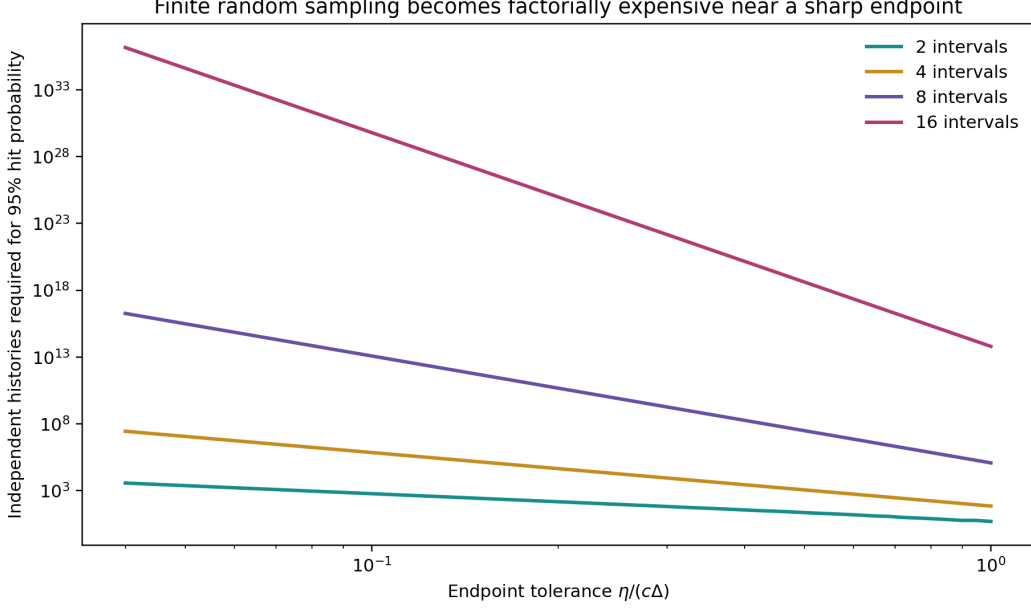


Figure 3: Exact number of independent matched-sampler histories needed for a 95% chance of approaching the maximal-loss endpoint. The cost grows factorially with the number of independently sampled intervals.

For N independent sampled histories,

$$\Pr\left(\min_{1 \leq r \leq N} D_r \leq \eta\right) = 1 - (1 - p_\eta)^N, \quad (38)$$

and the smallest N giving probability at least γ is

$$N_{\min} = \left\lceil \frac{\log(1 - \gamma)}{\log(1 - p_\eta)} \right\rceil. \quad (39)$$

Proof. The variables $(c - \varepsilon_j)/c$ are independent $\text{Uniform}(0, 1)$. Exact endpoint attainment has probability zero. For $\eta/(c\Delta) \leq 1$, the event that their sum is at most $\eta/(c\Delta)$ is an m -simplex of volume $[\eta/(c\Delta)]^m/m!$ inside the unit cube. Equations (38)–(39) follow from independent order statistics. $\square$

With eight intervals and relative tolerance $\eta/(c\Delta) = 0.2$, a 95% hit probability requires 47,182,783,307 independent histories. With 16 intervals it requires about 9.56×10^{24} . This theorem concerns the transparent matched sampler only. Actual samplers can concentrate differently, but no finite random sample proves an endpoint unless its proposal mechanism supplies a corresponding coverage guarantee. The next section therefore evaluates CRABS directly instead of treating this analytic example as a software benchmark.

7 Prospectively frozen comparison with CRABS

7.1 Design and decision rules

The confirmatory protocol was locally frozen before execution and before inspecting any confirmatory result; it was not registered in an external repository. The comparison uses the explicit bridge $\rho = 1$, $\lambda_{\text{ref}} = \lambda_p$ and $\mu_{\text{ref}} = 0$ on identical piecewise-linear time grids. CRABS's `p.delta` proposal parameter is recorded separately from the pulled signal λ_p and is never substituted for it.

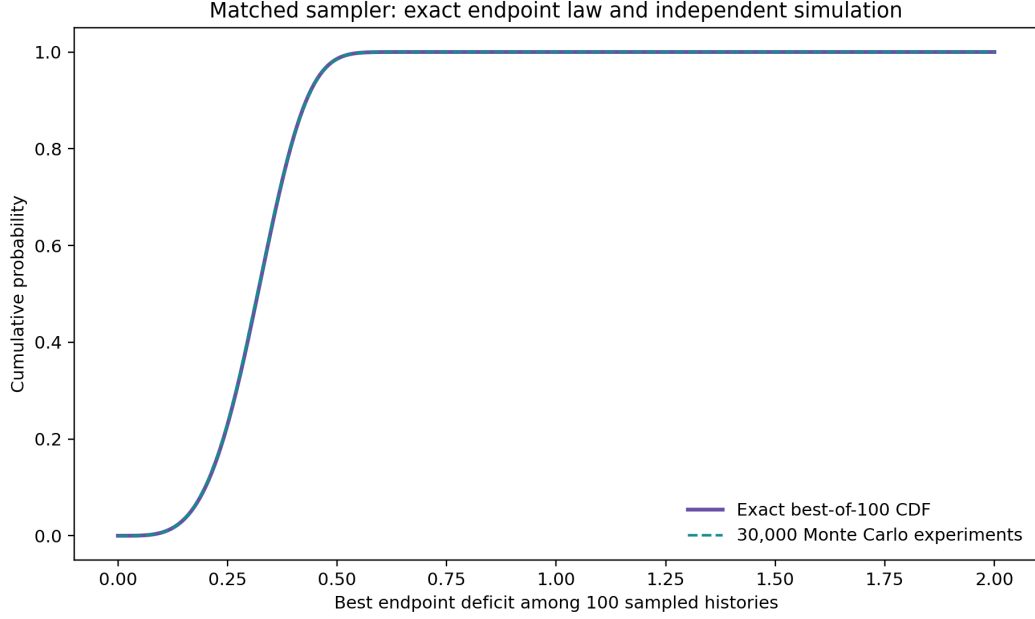


Figure 4: Exact best-of-100 endpoint-deficit distribution for four independently sampled intervals, compared with 30,000 direct Monte Carlo experiments. This validates the order-statistic calculation; it is not a benchmark of the CRABS package.

The stochastic design contains 1,100 cells: five signals (constant, abrupt, oscillatory, high dynamic range and a qualified CRABS primates EBD signal), two grids (25 and 100 knots), and ten replicates. Two hundred cells exercise the native HSMRF/GMRF procedures. The other 900 are cap-matched cells at turnover caps $c \in \{0.25, 0.50, 0.90\}$ using CRABS rejection exploration, a cap-aware proposal control and an exact boundary constructor. Seeds, proposal ceilings, file prefixes, query ages, query multipliers and decision thresholds were fixed in the protocol.

The primary H2 analysis comprises 300 CRABS rejection cells. A cell satisfies H2 if the returned cloud leaves normalized deficit greater than 0.05 at one or more sharp discrete endpoints, or if a prospectively defined structural censor prevents a returned finite cloud. H3 requires the boundary constructor to reach both discrete endpoints in all 300 matched cells. H4 evaluates four ages crossed with four multipliers in every non-abrupt rejection cell: certification must change at least 20% of 3,840 finite-sampler query statuses and produce no false certificate. H5 requires clean installation, agreement under normal and optimized Python tasks, and median exact-certification time below the frozen two-second allowance.

7.2 Nontermination incident and complete accounting

The first bounded execution exposed an internal unbounded proposal loop in abrupt-signal cells. For those cells, the exact minimum cap can require $\lambda > 2$ while the frozen CRABS setting used `max.lambda=2`. A documented post-incident amendment, made before the complete rerun and before any primary H2 or H4 rejection result was available, added a deterministic preflight classification for this incompatibility. It changed no seed, signal, grid, cap, decision threshold, query or replacement rule.

The completed ledger accounts for all 1,100 planned cells and contains 800 hash-bound sidecars. It records 1,000 passes and 100 `CENSORED_STRUCTURAL_NONTERMINATION` outcomes: 40 native HSMRF/GMRF abrupt cells and 60 primary CRABS-rejection abrupt cells. A structural censor is an integrity-preserving observation about the frozen execution configuration, not evidence that finite sampling succeeded and not a returned-cloud endpoint estimate.

7.3 Confirmatory results

Gate	Target	Sealed result	Verdict
H1	Deterministic validity	1,944/1,944 validation cells passed	Pass
H2	Endpoint incompleteness	300/300 primary cells met the deficit-or-censor event; descriptive Wilson 95% range 98.74–100%	Pass
H3	Constructive attainability	Both discrete endpoints attained in 300/300 boundary cells; zero failures	Pass
H4	Decision utility	532/3,840 statuses changed (13.85%; descriptive Wilson 95% range 12.80–14.98%), below the 20% gate; zero false certificates	Fail
H5	Operational viability	Clean install and normal/optimized tasks passed; median certification time approximately 4.68×10^{-5} seconds versus two seconds allowed	Pass

Table 1: Prospectively frozen confirmatory gates. Wilson intervals are descriptive summaries, not replacements for the fixed pass/fail rules. H2 includes 240 returned finite clouds and 60 prospectively classified structural censors.

H2 and H4 answer different questions. H2 shows that the frozen finite exploration did not certify sharp endpoints: every evaluable returned cloud missed at least one endpoint beyond tolerance, and the remaining abrupt cells were structurally censored. H4 asks whether exact certification changed the particular decisions fixed in advance. It did so for 532 queries, but the pooled 13.85% fraction did not reach the 20% criterion. A discrete-grid sensitivity analysis changed 182 of 3,840 statuses (4.74%; descriptive Wilson range 4.11–5.46%). No analysis produced a false certificate.

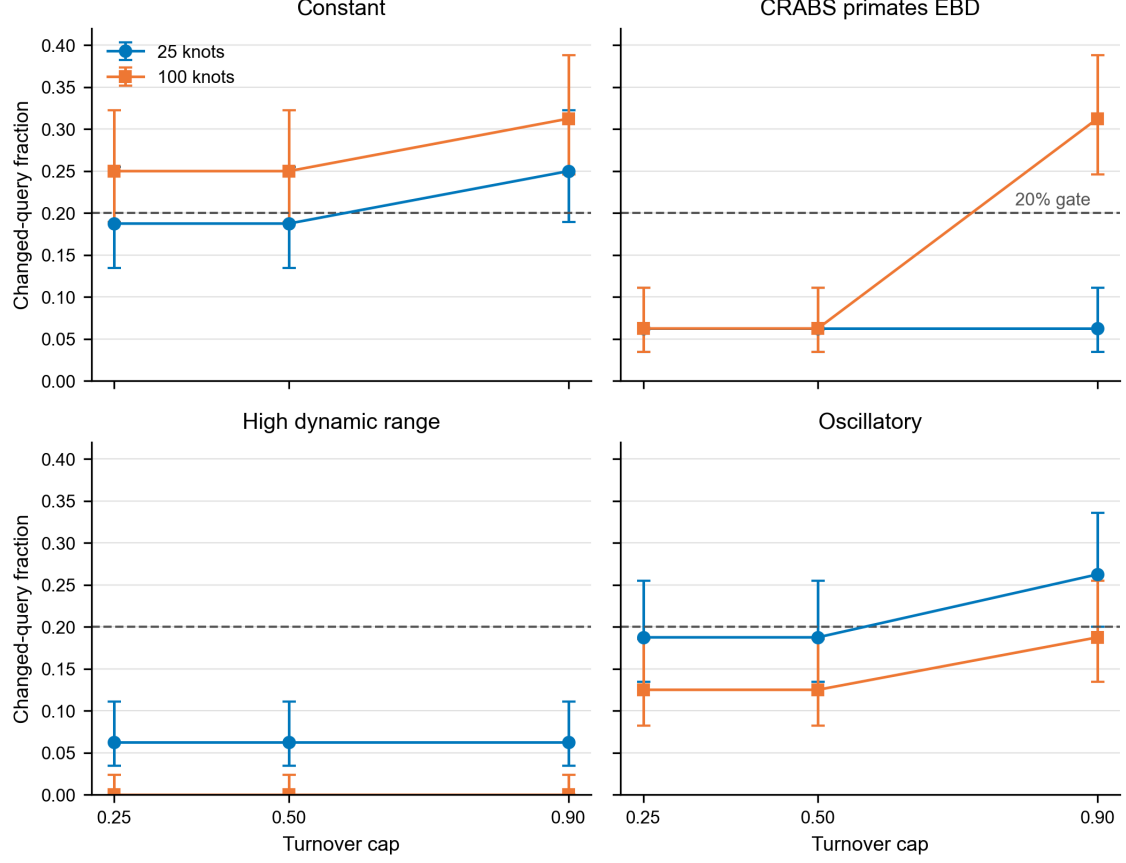


Figure 5: H4 changed-query fractions in the 24 evaluable signal-grid-cap strata, with descriptive Wilson intervals and the frozen 20% reference line. Each point aggregates 160 queries. Abrupt-signal strata are absent because all 60 primary rejection cells were structural censors and therefore had no finite-cloud decisions to compare. Some individual strata cross 20%, but the prospectively specified decision is pooled; the subgroup pattern does not rescue the failed H4 gate.

The direct comparison therefore supports exact certification as a fast diagnostic safeguard against endpoint incompleteness. It does not support the stronger claim that certification is an empirically essential or “must-have” complement to CRABS under this benchmark.

8 External deterministic constraints

Let

$$M(\tau) = \frac{M_0}{F(\tau)} \quad (40)$$

be the model-defined pulled lineage trajectory anchored at M_0 at the analysis origin. Define the associated deterministic diversity trajectory

$$N_{\text{det}}(\tau) = \frac{M(\tau)}{u(\tau)} = \frac{M_0}{q(\tau)}. \quad (41)$$

This is a deterministic model quantity. It is not, without an observation model, the realised species richness of a historical clade.

A supplied model-scale statement $N_{\text{det}}(\tau_i) \geq D_i$ is equivalent to

$$q(\tau_i) \leq \frac{M_0}{D_i}, \quad n(x_i) \geq x_i - \frac{M_0}{D_i}. \quad (42)$$

Corollary 13 (Minimum cap implied by one deterministic diversity bound). *With no interior event between the analysis origin and $x_i > 1$, the minimum cap compatible with (42) is*

$$c_{\min} = \left[1 - \frac{M_0/D_i - \rho}{x_i - 1} \right]_+. \quad (43)$$

For a shallower scale $x < x_i$, the same constraint implies

$$n(x) \geq x_i - \frac{M_0}{D_i} - c(x_i - x), \quad (44)$$

so information at an older age propagates towards the analysis origin.

The algebra is exact for N_{det} . Applying it to fossil observations requires three further bridges: from the deterministic trajectory to a realised latent richness process, from preserved occurrences to latent richness, and from genera or higher taxa to the species scale. An incompatibility therefore falsifies the cap and those bridge assumptions jointly, not the cap alone.

9 Plug-in mammal sensitivity illustration

The public data and code of Upham et al. (2021) include pulled-rate estimates for mammal timetrees and fossil-genus summaries. We use one extracted trajectory only. The source tree was sliced at 1 Ma, so the analysis origin contains $M_0 = 4,790$ lineages; this is not present-day mammal diversity. The calculation holds F fixed and supplies no coverage claim.

At actual age 58.5 Ma ($\tau = 57.5$ after the shifted origin), the extracted values are

$$\lambda_p = 0.0438809486 \text{ Myr}^{-1}, \quad F = 119.7090495. \quad (45)$$

Under $c = 0.5$ and no external deterministic constraint,

$$0.04388095 \leq \lambda \leq 0.08703484,$$

a factor of 1.98343. Treating the cross-bin value $D = 44$ as a purely illustrative lower bound changes the interval to

$$0.04825254 \leq \lambda \leq 0.08703484,$$

a factor of 1.80374. Values $D = 398$ and $D = 577$ are incompatible with $c = 0.5$ under the full deterministic and taxonomic mapping; their minimum compatible caps are 0.90704 and 0.93849. These numbers diagnose assumptions. They do not estimate mammalian turnover or validate a genus-to-species bridge.

10 Conditional geometry versus statistical uncertainty

Let $\Theta(F)$ denote the conditional functional identified set generated by a pulled-scale trajectory F . Suppose a simultaneous random set $\mathcal{F}_{1-\alpha}$ satisfies

$$\Pr(F_0 \in \mathcal{F}_{1-\alpha}) \geq 1 - \alpha. \quad (46)$$

Then

$$\Theta(F_0) \subseteq \bigcup_{F \in \mathcal{F}_{1-\alpha}} \Theta(F) \quad \text{whenever } F_0 \in \mathcal{F}_{1-\alpha}, \quad (47)$$

so the union transfers the simultaneous coverage of the signal set to a set containing the true conditional identified set. This is a set-inclusion principle, not an operational inferential method. A pointwise band is not automatically simultaneous; this release does not construct $\mathcal{F}_{1-\alpha}$ or propagate phylogenetic, smoothing and model uncertainty.

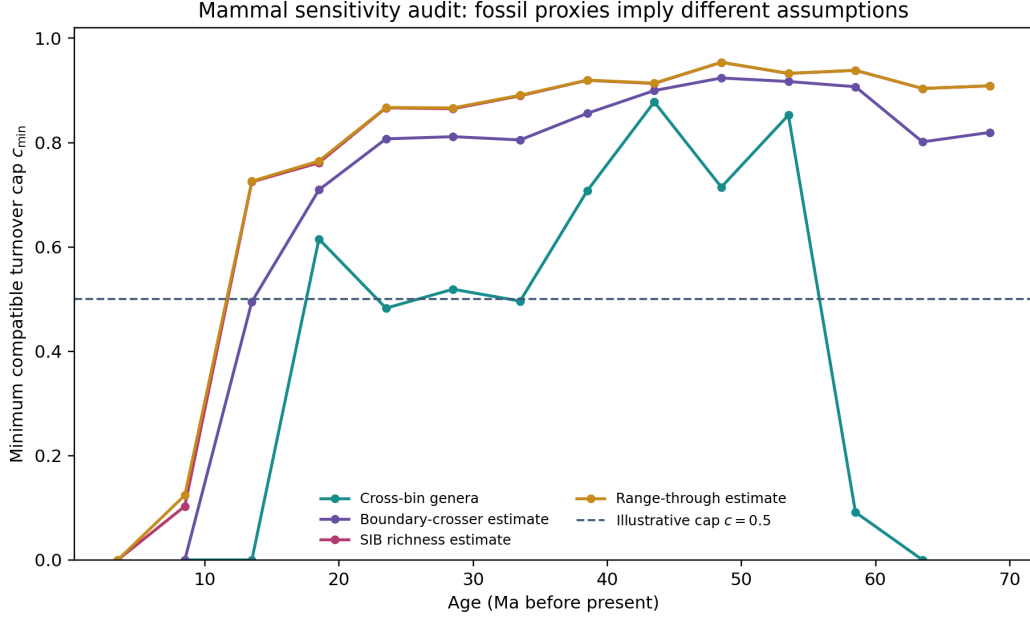


Figure 6: Single-tree plug-in sensitivity with no simultaneous signal band or fossil observation model. Different fossil-genus summaries imply different minimum caps only under the stated deterministic and taxonomic mappings.

11 Position relative to existing methods

Question	Louca–Pennell	Congruence exploration	Identifiable restricted or enriched models	Present candidate
What is learned?	Pulled signal and congruence	Constructed or sampled histories; shared patterns	Point identification within stronger model or data classes	Conditional functional set and target projections
Endpoint guarantee	No	Proposal-dependent finite exploration	Point estimate under model	Closed-form sharp endpoints for stated classes
Infeasibility witness	No	Not the principal output	Model fit or identification result	Violating pair and minimum cap
Deterministic events	Process literature required	Event scenarios may be constructed	Model-specific	Stieltjes coordinate and fixed-stem proof
Pulled-signal uncertainty	Estimation problem remains	Method-specific	Method-specific	Not supplied; union architecture only

Table 2: The proposed method complements rather than supersedes existing approaches. Congruence tools explore histories; identifiable restrictions or richer observations can restore point identification (Legried and Terhorst 2022; Legried and Terhorst 2023; Truman et al. 2025; Truman et al. 2026; Dieselhorst and Stadler 2026).

The integrating factor, pulled scale and zero-inflated geometric branching laws have reconstructed-process antecedents (Lambert and Stadler 2013; Höhna 2015; MacPherson et al. 2022). Partial-identification framing, generic Lipschitz extensions and infinite-dimensional linear programming are also established literatures (Manski 2003; McShane 1934; Whitney 1934; Anderson and Nash 1987). The defensible priority claim is narrower: we found no prior work coupling this cumulative-loss measure fibre to the complete package of cap-regime bounds, finite certificates, barrier test, minimum-cap diagnostic and exact endpoint-sampling law. The search was targeted, not systematic. A specialist recognition check remains necessary.

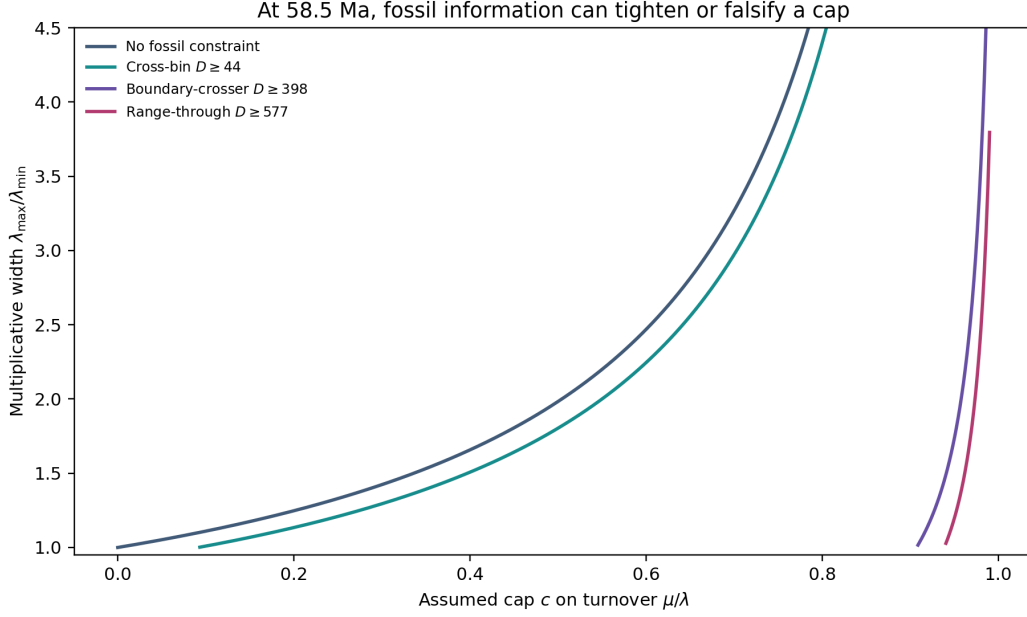


Figure 7: Worked plug-in sensitivity at 58.5 Ma. Curves report conditional mathematical compatibility, not confidence intervals or realised palaeodiversity.

12 Computational assurance

The theorem package inherited from release 0.2.1 includes:

- 17 deterministic unit tests covering the smooth chart, cap regimes, event algebra, fixed-stem normalisation, finite certificates, boundary validation, mammal replay and sampling theorem;
- an independently written loop-based envelope implementation and 720 linear-programme endpoint checks across 180 random feasible cases;
- a direct 160,000-replicate birth–death simulation with two deterministic bottlenecks, checked against analytic survival and the conditional descendant-count law;
- five deliberately broken or infeasible negative controls, all detected;
- exact dependency records, component-level licences, claim-to-evidence mappings and scoped assurance states;
- machine-readable data and alt text for every figure, plus PDF render inspection.

These checks establish internal replay and independent numerical consistency. They do not constitute external mathematical review, biological validation or formal proof checking. Exact dependencies are recorded, and the complete check suite passes under normal and optimized Python in a clean container pinned to a Python 3.13.5 base-image digest. That recreation was performed by the release-preparing agent, so independent external reproduction remains unassessed.

The 0.3.0-candidate review branch adds a complete 1,100-cell stochastic ledger, 800 hash-bound sidecars, an immutable full-ledger manifest, frozen hypothesis summaries, dual-mode protocol controls, seven protocol-harness tests and a clean-install/runtime receipt. The H5 record preserves an initial environment-selection incident in which a Python interpreter lacked SciPy and failed before unit-test execution; the additive corrected receipt changes only interpreter

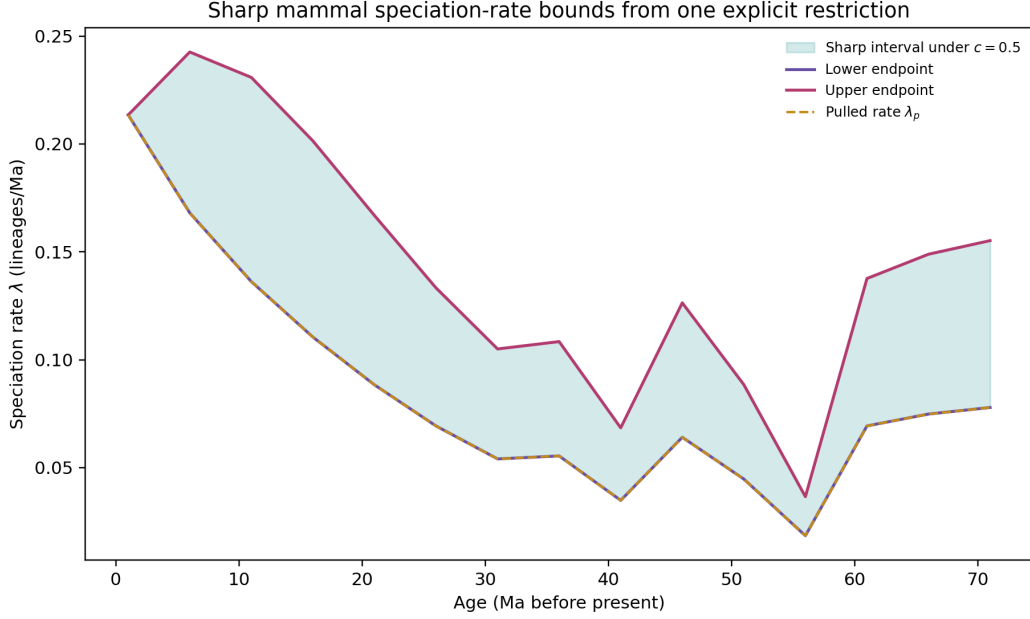


Figure 8: Sharp pointwise speciation bounds under $c = 0.5$ for one fixed mammal pulled-rate curve. The restriction excludes continuous negative net diversification; the figure contains no pulled-signal uncertainty.

qualification. The existing 0.2.1 release manifest is intentionally not rewritten to absorb the new evidence.

13 Limitations and research programme

The review draft remains a candidate for five reasons. First, the fixed-stem theorem needs an external specialist audit and does not cover crown or random-origin conditioning. Second, statistical inference requires a simultaneous uncertainty set for the pulled signal. Third, fossil application requires a preservation, observation and taxonomic mapping model. Fourth, the confirmatory comparison is bounded to one explicit semantic bridge, five signals, two grids, three caps and preset decision queries; H4 failed, and structural censoring limits returned-cloud claims for abrupt signals. Fifth, priority should be checked systematically across stochastic-process, survival-analysis, viability, optimal-recovery and non-English literatures.

The most valuable extensions are: exact robust envelopes over a simultaneous family of F curves; observation-aware fossil inequalities or chance constraints; target-specific optimisation for peak timing and integrated extinction; prospectively chosen target-specific decisions; sampler endpoint guarantees; crown-conditioned event laws; and formalisation of Theorems 1, 6 and 8 in a proof assistant. New tuning choices, signals or thresholds may motivate a separate study but cannot retroactively repair the failed H4 gate.

14 Conclusion

A congruence class is a functional identified set, not a collection of curves returned by one sampler. The cumulative-loss coordinate makes a useful part of that set affine. Under explicit restrictions, the analyst can report feasible trajectories, exact target endpoints, endpoint-attaining histories and short incompatibility certificates. In the frozen CRABS comparison, finite rejection exploration missed an endpoint tolerance in every primary cell under the prespecified deficit-or-censor rule, yet certification changed only 13.85% of the evaluable decision queries and

therefore failed the 20% utility gate. The evidence supports a fast exact diagnostic, not a claim of empirical necessity. Fixed-signal geometry, finite exploration, statistical uncertainty and biological observation models remain separate layers.

A Sanity checks

Complete-sampling pure birth. With $\nu = 0$, $A = 0$, $u = 1$, $q = F$, $\lambda = \lambda_p$ and $\mu = 0$. With incomplete sampling but no extinction, $\nu = (1 - \rho)\delta_0$.

Constant turnover. If $\rho = 1$ and $d\nu = c dF$, then

$$A = c(F - 1), \quad q = 1 + (1 - c)(F - 1), \quad \lambda = \frac{\lambda_p F}{1 + (1 - c)(F - 1)},$$

and $d\nu/dF = c = \mu/\lambda$.

Instantaneous event. For $m = (1 - s)q(a-)$,

$$\frac{u(a+)}{u(a-)} = \frac{q(a+)}{q(a-)} = s$$

exactly.

Fixed-stem normalisation. For n sampled descendants, integrating (19) over the ordered simplex gives

$$\frac{u(T)}{F(T)} \left(1 - \frac{1}{F(T)}\right)^{n-1} = p_n(T),$$

which checks both the topology-marginal branching-time multiplicity factor and the event-invariant p_1/u ratio.

Endpoint sampling. For $m = 1$, (37) reduces to the uniform distribution on $[0, c\Delta]$. For $m = 2$, it gives the triangular lower tail $\eta^2/[2(c\Delta)^2]$.

B Reproduction commands

From the release root:

```
make verify
make verify-package
docker build -f Containerfile -t affine-diversification-fibres:0.2.1 .
docker run --rm affine-diversification-fibres:0.2.1
stage2=experiments/affine_crabs_confirmatory/execution
shasum -a 256 -c "$stage2/STOCHASTIC_STAGE2_MANIFEST.sha256"
python experiments/affine_crabs_confirmatory/generate_review_figure.py
```

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